Enterprise Access Point & Centralized DHCP Relay Simulation

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Tool Used:
Cisco Packet Tracer

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PROJECT LABORATORY OVERVIEW

This Cisco Packet Tracer simulation designed to showcase hands-on proficiency in enterprise network infrastructure, Layer 2/3 traffic management, and practical network troubleshooting.

The primary objective of this laboratory is to simulate centralized IP address distribution across multiple segmented VLANs using a single central DHCP Server via ip helper-address (DHCP Relay) over an enterprise MDF?IDF switch hierarchy.

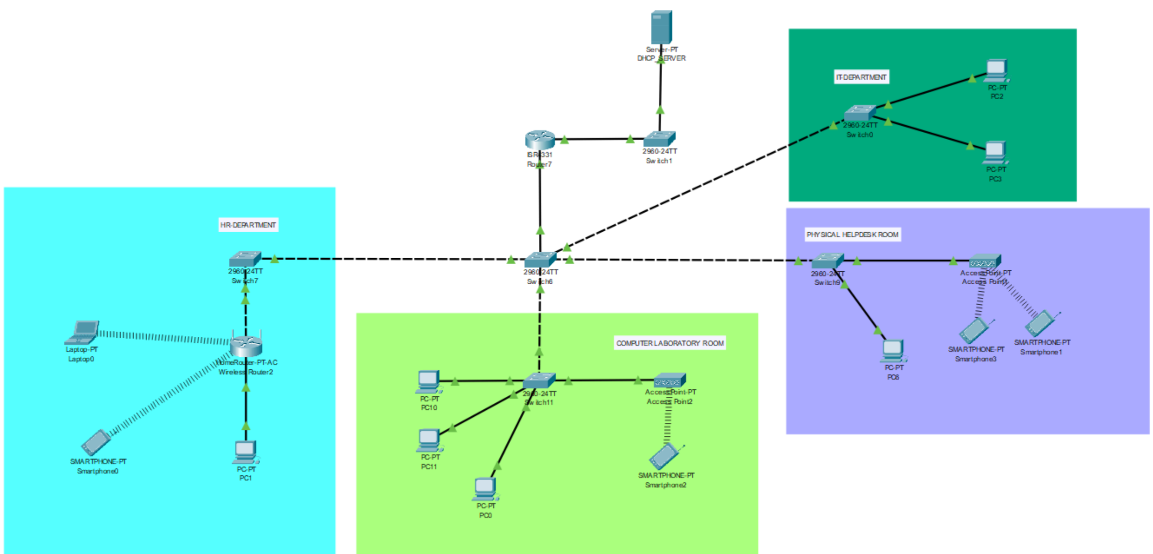


Figure 1: DHCP Relay Topology

The topology models a multi-department enterprise network using a standard Main Distribution Frame (MDF) and Intermediate Distribution Frame (IDF) hierarchy. It also demonstrates core networking concepts such as VLAN segmentation, VLSM/CIDR Subnetting, inter-VLAN routing, DHCP, layer 2 security mitigation, and real-world problem scenarios (break/fix).

Network Architecture and Hardware Roles

The topology models a multi-department enterprise network using a standard Main Distribution Frame (MDF) and Intermediate Distribution Frame (IDF) hierarchy.

Device Name	Device Model / Type	Network Role & Function
MAIN_ROUTER	Cisco ISR 4331/K9	Central Layer 3 Gateway performing Router-on-a-Stick (ROAS) Inter-VLAN routing and acting as the DHCP Relay Agent via IP Helper-Address
BRIDGE_SWITCH_1	Cisco Switch	Layer 2 bridge link connecting MAIN_ROUTER directly to the central DHCP Server (Server0).
MDF_SWITCH (Switch 6)	Cisco 2960 Switch	Main Distribution Frame (MDF) core switch managing primary trunk connections across all departments.
HR_SWITCH (Switch 7)	Cisco 2960 Switch	IDF servicing the Human Resources Department (VLAN 20).
COMLAB_SWITCH (Switch 11)	Cisco 2960 Switch	IDF servicing the Computer Laboratory Room (VLAN 10).
HELPDESK_SWITCH (Switch 9)	Cisco 2960 Switch	IDF servicing the Physical Helpdesk Room (VLAN 30).

ITDEPT_SWITCH (Switch 0)	Cisco 2960 Switch	IDF servicing the IT Department (VLAN 50).
Guest AP (AP 1)	Lightweight Access Point	Wireless Access Point broadcasting Guest Wi-Fi (VLAN 40).
ComLab AP (AP 2)	Lightweight Access Point	Wireless Access Point broadcasting Computer Lab Wi-Fi (VLAN 10).

IP Subnetting Calculations

The laboratory consists of multiple department VLANs, Planned Capacity, and the Allocated CIDR Block Size to each department.

Given Base Network: 192.168.10.0/24

Department / VLAN	Planned Capacity	Allocated CIDR Block Size
VLAN 10 (COMLAB)	25 - 30 Hosts	/27 (30 Hosts)
VLAN 20 (HR_DEPT)	10 - 14 Hosts	/28 (14 Hosts)
VLAN 30 (IT_DEPT)	10 - 14 Hosts	/28 (14 Hosts)
VLAN 40 (GUEST_WIFI)	10 - 14 Hosts	/28 (14 Hosts)
VLAN 50 (SERVERS)	4-6 Hosts	/29 (6 Hosts)
VLAN 100 (UNUSED_PORTS)	–	–

Assumed planned hosts for each department:

- Computer Laboratory Room - 30 hosts
- HR Department - 14 hosts
- IT Department - 14 hosts
- Helpdesk Room (for guest wifi) - 10 hosts
- Helpdesk Room (company's devices) - 6 hosts
- Servers - 4 hosts

IP Subnetting Calculations:

1st subnet/VLAN 10 (computer laboratory room):

- Network: 192.168.10.0/27
- Hosts: 192.168.10.1 - 192.168.10.30 (/27)
- Broadcast: 192.168.10.31/27

2nd Subnet/VLAN 20 (HR Department):

- Network: 192.168.10.32/28
- Hosts: 192.168.10.33 - 192.168.10.46 (/28)
- Broadcast: 192.168.10.47/28

3rd Subnet/VLAN 30 (IT Department):

- Network: 192.168.10.48/28
- Hosts: 192.168.10.49 - 192.168.10.62 (/28)
- Broadcast: 192.168.10.63/28

4th Subnet/VLAN 40 (Helpdesk Room - guest wifi):

- Network: 192.168.10.64/28
- Hosts: 192.168.10.65 - 192.168.10.78 (/28)
- Broadcast: 192.168.10.79/28

5th Subnet/VLAN 50 (Server):

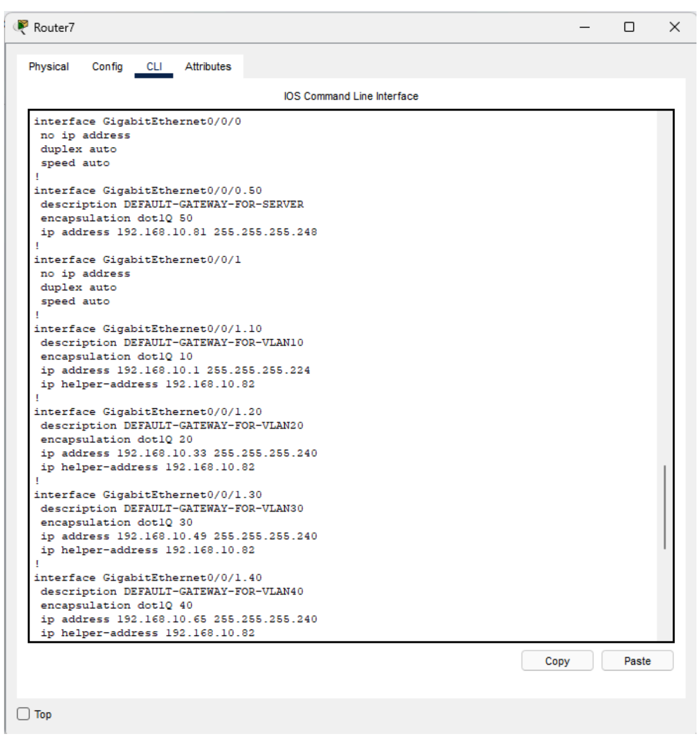
- Network: 192.168.10.80/29
- Hosts: 192.168.10.81 - 192.168.10.86 (/29)
- Broadcast: 192.168.10.87/29

CONFIGURATIONS

This section aims to showcase and demonstrate the necessary configurations made throughout this entire laboratory set up.

Main Router Configuration

1. **Cisco ISR4331/K9 configuration (Router 7 - MAIN_ROUTER):**
    - **Purpose:** Acting as a centralized router on a stick, inter-VLAN, relaying IP requests via IP helper-address from the DHCP server to the end devices and vice versa.
    - **Configurations:**
        - Configured GigabitEthernet0/0/0.50 interface as VLAN 50, serving as a default gateway pathway for servers.
        - On the otherhand, GigabitEthernet0/0/1 was used as an interface for VLAN 10, 20, 30, and 40.
    - **Running-configuration Screenshot:**




```
interface GigabitEthernet0/0/0
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/0.50
 description DEFAULT-GATEWAY-FOR-SERVER
 encapsulation dot1Q 50
 ip address 192.168.10.81 255.255.255.248
!
interface GigabitEthernet0/0/1
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/1.10
 description DEFAULT-GATEWAY-FOR-VLAN10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.224
 ip helper-address 192.168.10.82
!
interface GigabitEthernet0/0/1.20
 description DEFAULT-GATEWAY-FOR-VLAN20
 encapsulation dot1Q 20
 ip address 192.168.10.33 255.255.255.240
 ip helper-address 192.168.10.82
!
interface GigabitEthernet0/0/1.30
 description DEFAULT-GATEWAY-FOR-VLAN30
 encapsulation dot1Q 30
 ip address 192.168.10.49 255.255.255.240
 ip helper-address 192.168.10.82
!
interface GigabitEthernet0/0/1.40
 description DEFAULT-GATEWAY-FOR-VLAN40
 encapsulation dot1Q 40
 ip address 192.168.10.65 255.255.255.240
 ip helper-address 192.168.10.82
```

Switch Configuration

1. **Configurations created for each switch:**
    - Created 6 VLANs:
        - VLAN 10 - For computer laboratory
        - VLAN 20 - For HR Department
        - VLAN 30 - For IT Department
        - VLAN 40 - For Guest WIFI
        - VLAN 50 - For Servers
        - VLAN 100 - For unused ports

2. **CONFIGURATION FOR SWITCH 6 (MDF_SWITCH):**
    - **Purpose:** Acting as the main distribution frame for all the switches.
        - **Configurations Created:**
            - Int range Fa0/1 - Fa0/5 are configured as a trunk.
            - Int range Fa0/6 - fa0/24, and int range g0/1 - 0/2 are configured as shutdown and placed to unused port VLAN.
    - **Configurations Screenshots:**

```
MDF_SWITCH#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   COMLAB                           active
20   HR_DEPT                          active
30   IT_DEPT                          active
40   GUEST_WIFI                       active
50   SERVERS                          active
100  UNUSED_PORTS                     active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
MDF_SWITCH#
```

```
MDF_SWITCH#sh int tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1
Fa0/3       on           802.1q         trunking      1
Fa0/4       on           802.1q         trunking      1
Fa0/5       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005
Fa0/2       1-1005
Fa0/3       1-1005
Fa0/4       1-1005
Fa0/5       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30,40,50,100
Fa0/2       1,10,20,30,40,50,100
Fa0/3       1,10,20,30,40,50,100
Fa0/4       1,10,20,30,40,50,100
Fa0/5       1,10,20,30,40,50,100

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30,40,50,100
Fa0/2       1,10,20,30,40,50,100
Fa0/3       1,10,20,30,40,50,100
Fa0/4       1,10,20,30,40,50,100
Fa0/5       1,10,20,30,40,50,100
```

```
MDF_SWITCH#sh int status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    trunk      auto    auto  10/100BaseTX
Fa0/2                        connected    trunk      auto    auto  10/100BaseTX
Fa0/3                        connected    trunk      auto    auto  10/100BaseTX
Fa0/4                        connected    trunk      auto    auto  10/100BaseTX
Fa0/5                        connected    trunk      auto    auto  10/100BaseTX
Fa0/6                        disabled 100        auto    auto  10/100BaseTX
Fa0/7                        disabled 100        auto    auto  10/100BaseTX
Fa0/8                        disabled 100        auto    auto  10/100BaseTX
Fa0/9                        disabled 100        auto    auto  10/100BaseTX
Fa0/10                       disabled 100        auto    auto  10/100BaseTX
Fa0/11                       disabled 100        auto    auto  10/100BaseTX
Fa0/12                       disabled 100        auto    auto  10/100BaseTX
Fa0/13                       disabled 100        auto    auto  10/100BaseTX
Fa0/14                       disabled 100        auto    auto  10/100BaseTX
Fa0/15                       disabled 100        auto    auto  10/100BaseTX
Fa0/16                       disabled 100        auto    auto  10/100BaseTX
Fa0/17                       disabled 100        auto    auto  10/100BaseTX
Fa0/18                       disabled 100        auto    auto  10/100BaseTX
Fa0/19                       disabled 100        auto    auto  10/100BaseTX
Fa0/20                       disabled 100        auto    auto  10/100BaseTX
Fa0/21                       disabled 100        auto    auto  10/100BaseTX
Fa0/22                       disabled 100        auto    auto  10/100BaseTX
Fa0/23                       disabled 100        auto    auto  10/100BaseTX
Fa0/24                       disabled 100        auto    auto  10/100BaseTX
Gig0/1                       disabled 100        auto    auto  10/100BaseTX
Gig0/2                       disabled 100        auto    auto  10/100BaseTX
```

3. CONFIGURATION FOR SWITCH 1 (BRIDGE_SWITCH_1):

- **Purpose:** Acting as the bridge between cisco ISR4331/K9 router to the server and vice versa.
- **Configurations Created:**
  - Int fa0/2 configured as a trunk.
  - Int fa0/1 configured as VLAN 50.
  - Int range Fa0/3 - fa0/24, and int range g0/1 - 0/2 are configured as shutdown and placed to unused port VLAN.
- **Configurations Screenshots:**

```
BRIDGE_SWITCH_1#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   COMLAB                           active
20   HR_DEPT                          active
30   IT_DEPT                          active
40   GUEST_WIFI                       active
50   SERVERS                          active    Fa0/1
100  UNUSED_PORTS                     active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
BRIDGE_SWITCH_1#
```

```
BRIDGE_SWITCH_1#sh int tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/2       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/2       1-1005

Port        Vlans allowed and active in management domain
Fa0/2       1,10,20,30,40,50,100

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/2       1,10,20,30,40,50,100

BRIDGE_SWITCH_1#
```

```
BRIDGE_SWITCH_1#sh int status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    50         auto    auto  10/100BaseTX
Fa0/2                        connected    trunk      auto    auto  10/100BaseTX
Fa0/3                        disabled 100        auto    auto  10/100BaseTX
Fa0/4                        disabled 100        auto    auto  10/100BaseTX
Fa0/5                        disabled 100        auto    auto  10/100BaseTX
Fa0/6                        disabled 100        auto    auto  10/100BaseTX
Fa0/7                        disabled 100        auto    auto  10/100BaseTX
Fa0/8                        disabled 100        auto    auto  10/100BaseTX
Fa0/9                        disabled 100        auto    auto  10/100BaseTX
Fa0/10                       disabled 100        auto    auto  10/100BaseTX
Fa0/11                       disabled 100        auto    auto  10/100BaseTX
Fa0/12                       disabled 100        auto    auto  10/100BaseTX
Fa0/13                       disabled 100        auto    auto  10/100BaseTX
Fa0/14                       disabled 100        auto    auto  10/100BaseTX
Fa0/15                       disabled 100        auto    auto  10/100BaseTX
Fa0/16                       disabled 100        auto    auto  10/100BaseTX
Fa0/17                       disabled 100        auto    auto  10/100BaseTX
Fa0/18                       disabled 100        auto    auto  10/100BaseTX
Fa0/19                       disabled 100        auto    auto  10/100BaseTX
Fa0/20                       disabled 100        auto    auto  10/100BaseTX
Fa0/21                       disabled 100        auto    auto  10/100BaseTX
Fa0/22                       disabled 100        auto    auto  10/100BaseTX
Fa0/23                       disabled 100        auto    auto  10/100BaseTX
Fa0/24                       disabled 100        auto    auto  10/100BaseTX
Gig0/1                       disabled 100        auto    auto  10/100BaseTX
Gig0/2                       disabled 100        auto    auto  10/100BaseTX
```

4. CONFIGURATION FOR SWITCH 0 (ITDEPT_SWITCH):

- **Purpose:** acting as an intermediate distribution frame for the IT department.
- **Configurations Created:**
  - Int fa0/1 configured as a trunk.
  - Int range fa0/2, fa0/3 configured as VLAN 30
  - Int range Fa0/4 - fa0/24, and int range g0/1 - 0/2 are configured as shutdown and placed to unused port VLAN.
- **Configurations Screenshots:**

```
ITDEPT_SWITCH#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   COMLAB                           active
20   HR_DEPT                          active
30   IT_DEPT                          active    Fa0/2, Fa0/3
40   GUEST_WIFI                       active
50   SERVERS                          active
100  UNUSED_PORTS                     active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
ITDEPT_SWITCH#
```

```
ITDEPT_SWITCH#sh int tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       auto         n-802.1q       trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30,40,50,100

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30,40,50,100

ITDEPT_SWITCH#
```

```
ITDEPT_SWITCH#sh int status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    trunk      auto    auto  10/100BaseTX
Fa0/2                        connected    30         auto    auto  10/100BaseTX
Fa0/3                        connected    30         auto    auto  10/100BaseTX
Fa0/4                        disabled 100        auto    auto  10/100BaseTX
Fa0/5                        disabled 100        auto    auto  10/100BaseTX
Fa0/6                        disabled 100        auto    auto  10/100BaseTX
Fa0/7                        disabled 100        auto    auto  10/100BaseTX
Fa0/8                        disabled 100        auto    auto  10/100BaseTX
Fa0/9                        disabled 100        auto    auto  10/100BaseTX
Fa0/10                       disabled 100        auto    auto  10/100BaseTX
Fa0/11                       disabled 100        auto    auto  10/100BaseTX
Fa0/12                       disabled 100        auto    auto  10/100BaseTX
Fa0/13                       disabled 100        auto    auto  10/100BaseTX
Fa0/14                       disabled 100        auto    auto  10/100BaseTX
Fa0/15                       disabled 100        auto    auto  10/100BaseTX
Fa0/16                       disabled 100        auto    auto  10/100BaseTX
Fa0/17                       disabled 100        auto    auto  10/100BaseTX
Fa0/18                       disabled 100        auto    auto  10/100BaseTX
Fa0/19                       disabled 100        auto    auto  10/100BaseTX
Fa0/20                       disabled 100        auto    auto  10/100BaseTX
Fa0/21                       disabled 100        auto    auto  10/100BaseTX
Fa0/22                       disabled 100        auto    auto  10/100BaseTX
Fa0/23                       disabled 100        auto    auto  10/100BaseTX
Fa0/24                       disabled 100        auto    auto  10/100BaseTX
Gig0/1                       disabled 100        auto    auto  10/100BaseTX
Gig0/2                       disabled 100        auto    auto  10/100BaseTX

ITDEPT_SWITCH#
```

5. CONFIGURATION FOR SWITCH 9 (HELPDESK_SWITCH):

- **Purpose:** acting as an intermediate distribution frame for the physical helpdesk room.
- **Configurations Created:**
  - Int fa0/1 configured as a trunk.
  - Int fa0/2 configured as VLAN 40
  - Int fa0/3 configured as VLAN 10 (part of computer laboratory devices)
  - Int range Fa0/4 - fa0/24, and int range g0/1 - 0/2 are configured as shutdown and placed to unused port VLAN.
- **Configurations Screenshots:**

```
HELPDESK_SWITCH#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   COMLAB                           active    Fa0/3
20   HR_DEPT                          active
30   IT_DEPT                          active
40   GUEST_WIFI                       active    Fa0/2
50   SERVERS                          active
100  UNUSED_PORTS                     active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
HELPDESK_SWITCH#
```

```
HELPDESK_SWITCH#sh int tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30,40,50,100

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30,40,50,100

HELPDESK_SWITCH#
```

```
HELPDESK_SWITCH#sh int status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    trunk      auto    auto  10/100BaseTX
Fa0/2                        connected    40         auto    auto  10/100BaseTX
Fa0/3                        connected    10         auto    auto  10/100BaseTX
Fa0/4                        disabled 100        auto    auto  10/100BaseTX
Fa0/5                        disabled 100        auto    auto  10/100BaseTX
Fa0/6                        disabled 100        auto    auto  10/100BaseTX
Fa0/7                        disabled 100        auto    auto  10/100BaseTX
Fa0/8                        disabled 100        auto    auto  10/100BaseTX
Fa0/9                        disabled 100        auto    auto  10/100BaseTX
Fa0/10                       disabled 100        auto    auto  10/100BaseTX
Fa0/11                       disabled 100        auto    auto  10/100BaseTX
Fa0/12                       disabled 100        auto    auto  10/100BaseTX
Fa0/13                       disabled 100        auto    auto  10/100BaseTX
Fa0/14                       disabled 100        auto    auto  10/100BaseTX
Fa0/15                       disabled 100        auto    auto  10/100BaseTX
Fa0/16                       disabled 100        auto    auto  10/100BaseTX
Fa0/17                       disabled 100        auto    auto  10/100BaseTX
Fa0/18                       disabled 100        auto    auto  10/100BaseTX
Fa0/19                       disabled 100        auto    auto  10/100BaseTX
Fa0/20                       disabled 100        auto    auto  10/100BaseTX
Fa0/21                       disabled 100        auto    auto  10/100BaseTX
Fa0/22                       disabled 100        auto    auto  10/100BaseTX
Fa0/23                       disabled 100        auto    auto  10/100BaseTX
Fa0/24                       disabled 100        auto    auto  10/100BaseTX
Gig0/1                       disabled 100        auto    auto  10/100BaseTX
Gig0/2                       disabled 100        auto    auto  10/100BaseTX

HELPDESK_SWITCH#
```

6. CONFIGURATION FOR SWITCH 11 (COMLAB_SWITCH):

- **Purpose:** acting as an intermediate distribution frame for the computer laboratory room.
- **Configurations Created:**
  - Int fa0/1 configured as a trunk.
  - Int range fa0/2, fa0/3, fa0/4, fa0/5 configured as VLAN 10
  - Int range Fa0/6 - fa0/24, and int range g0/1 - 0/2 are configured as shutdown and placed to unused port VLAN.
- **Configurations Screenshots:**

```
COMLAB_SWITCH#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   COMLAB                           active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
20   HR_DEPT                          active
30   IT_DEPT                          active
40   GUEST_WIFI                       active
50   SERVERS                          active
100  UNUSED_PORTS                     active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
COMLAB_SWITCH#
```

```
COMLAB_SWITCH#sh int tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30,40,50,100

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30,40,50,100

COMLAB_SWITCH#
```

```
COMLAB_SWITCH#sh int status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    trunk      auto    auto  10/100BaseTX
Fa0/2                        connected    10         auto    auto  10/100BaseTX
Fa0/3                        connected    10         auto    auto  10/100BaseTX
Fa0/4                        connected    10         auto    auto  10/100BaseTX
Fa0/5                        connected    10         auto    auto  10/100BaseTX
Fa0/6                        disabled 100        auto    auto  10/100BaseTX
Fa0/7                        disabled 100        auto    auto  10/100BaseTX
Fa0/8                        disabled 100        auto    auto  10/100BaseTX
Fa0/9                        disabled 100        auto    auto  10/100BaseTX
Fa0/10                       disabled 100        auto    auto  10/100BaseTX
Fa0/11                       disabled 100        auto    auto  10/100BaseTX
Fa0/12                       disabled 100        auto    auto  10/100BaseTX
Fa0/13                       disabled 100        auto    auto  10/100BaseTX
Fa0/14                       disabled 100        auto    auto  10/100BaseTX
Fa0/15                       disabled 100        auto    auto  10/100BaseTX
Fa0/16                       disabled 100        auto    auto  10/100BaseTX
Fa0/17                       disabled 100        auto    auto  10/100BaseTX
Fa0/18                       disabled 100        auto    auto  10/100BaseTX
Fa0/19                       disabled 100        auto    auto  10/100BaseTX
Fa0/20                       disabled 100        auto    auto  10/100BaseTX
Fa0/21                       disabled 100        auto    auto  10/100BaseTX
Fa0/22                       disabled 100        auto    auto  10/100BaseTX
Fa0/23                       disabled 100        auto    auto  10/100BaseTX
Fa0/24                       disabled 100        auto    auto  10/100BaseTX
Gig0/1                       disabled 100        auto    auto  10/100BaseTX
Gig0/2                       disabled 100        auto    auto  10/100BaseTX

COMLAB_SWITCH#
```

7. CONFIGURATION FOR SWITCH 7 (HR_SWITCH):

- **Purpose:** acting as an intermediate distribution frame for the HR Department.
- **Configurations Created:**
  - Int fa0/1 configured as a trunk.
  - Int fa0/2 configured as VLAN 20
  - Int range Fa0/3 - fa0/24, and int range g0/1 - 0/2 are configured as shutdown and placed to unused port VLAN.
- **Configurations Screenshots:**

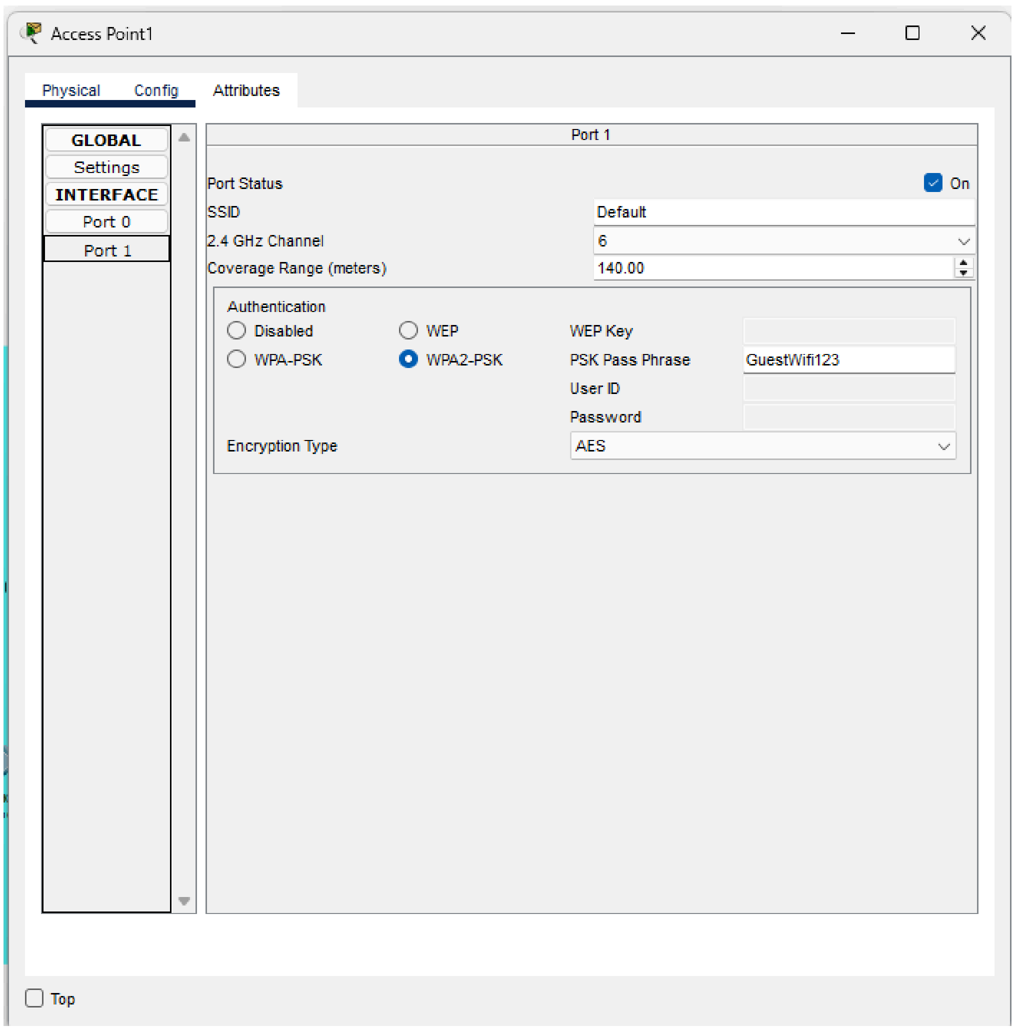
```
HR_SWITCH#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   COMLAB                           active
20   HR_DEPT                          active    Fa0/2
30   IT_DEPT                          active
40   GUEST_WIFI                       active
50   SERVERS                          active
100  UNUSED_PORTS                     active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
HR_SWITCH#
```

```
HR_SWITCH#sh int tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       auto         n-802.1q       trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30,40,50,100

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30,40,50,100

HR_SWITCH#
```

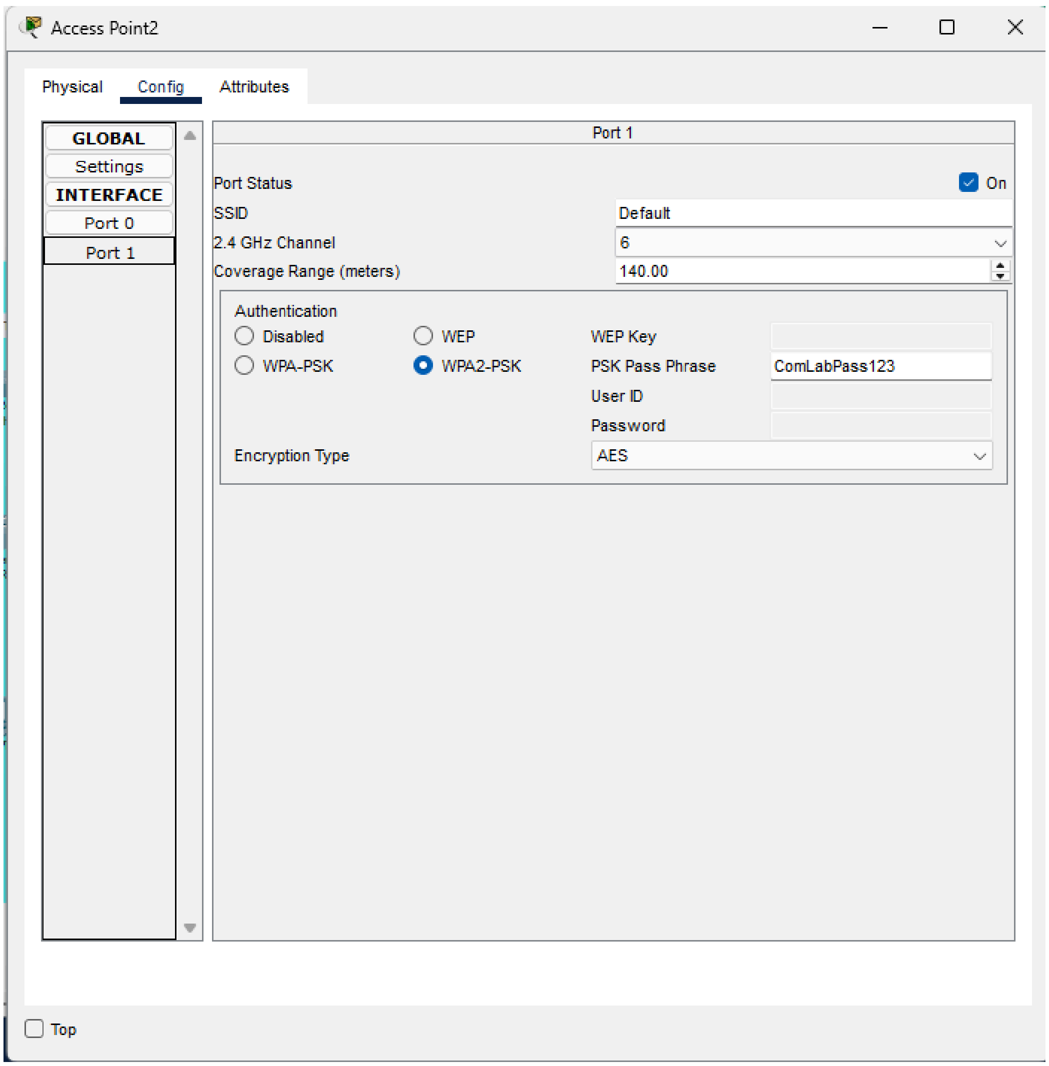
```
HR_SWITCH#sh int status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    trunk      auto    auto  10/100BaseTX
Fa0/2                        connected    20         auto    auto  10/100BaseTX
Fa0/3                        disabled 100        auto    auto  10/100BaseTX
Fa0/4                        disabled 100        auto    auto  10/100BaseTX
Fa0/5                        disabled 100        auto    auto  10/100BaseTX
Fa0/6                        disabled 100        auto    auto  10/100BaseTX
Fa0/7                        disabled 100        auto    auto  10/100BaseTX
Fa0/8                        disabled 100        auto    auto  10/100BaseTX
Fa0/9                        disabled 100        auto    auto  10/100BaseTX
Fa0/10                       disabled 100        auto    auto  10/100BaseTX
Fa0/11                       disabled 100        auto    auto  10/100BaseTX
Fa0/12                       disabled 100        auto    auto  10/100BaseTX
Fa0/13                       disabled 100        auto    auto  10/100BaseTX
Fa0/14                       disabled 100        auto    auto  10/100BaseTX
Fa0/15                       disabled 100        auto    auto  10/100BaseTX
Fa0/16                       disabled 100        auto    auto  10/100BaseTX
Fa0/17                       disabled 100        auto    auto  10/100BaseTX
Fa0/18                       disabled 100        auto    auto  10/100BaseTX
Fa0/19                       disabled 100        auto    auto  10/100BaseTX
Fa0/20                       disabled 100        auto    auto  10/100BaseTX
Fa0/21                       disabled 100        auto    auto  10/100BaseTX
Fa0/22                       disabled 100        auto    auto  10/100BaseTX
Fa0/23                       disabled 100        auto    auto  10/100BaseTX
Fa0/24                       disabled 100        auto    auto  10/100BaseTX
Gig0/1                       disabled 100        auto    auto  10/100BaseTX
Gig0/2                       disabled 100        auto    auto  10/100BaseTX

HR_SWITCH#
```

Access Point Configuration

1. **Access Point 1 Configuration (Guest Wi-Fi):**
    - **Purpose:** Acting as a wireless access point for guests.
    - **Configurations:**
        - Connected to fa0/2 interface of switch 9 (HELPDESK_SWITCH) as VLAN 40.
        - WPA2-PSK Security.
    - **Configurations Screenshot:**



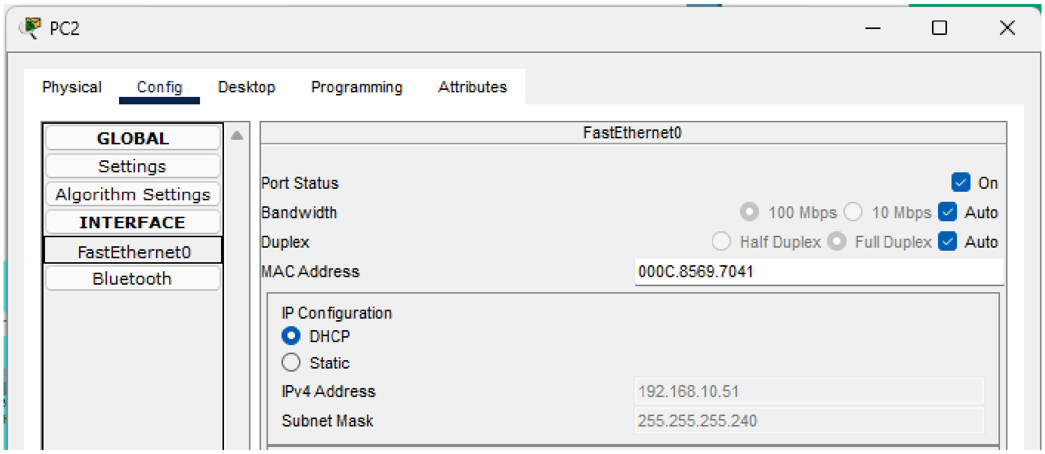
2. **Access Point 2 Configuration (COMLAB):**
    - **Purpose:** Acting as a wireless connection guests
    - **Configurations:**
        - Connected via fa0/2 interface of switch 11 (COMLAB_SWITCH) as VLAN 10.
        - WPA2-PSK Security.
    - **Configurations Screenshot:**

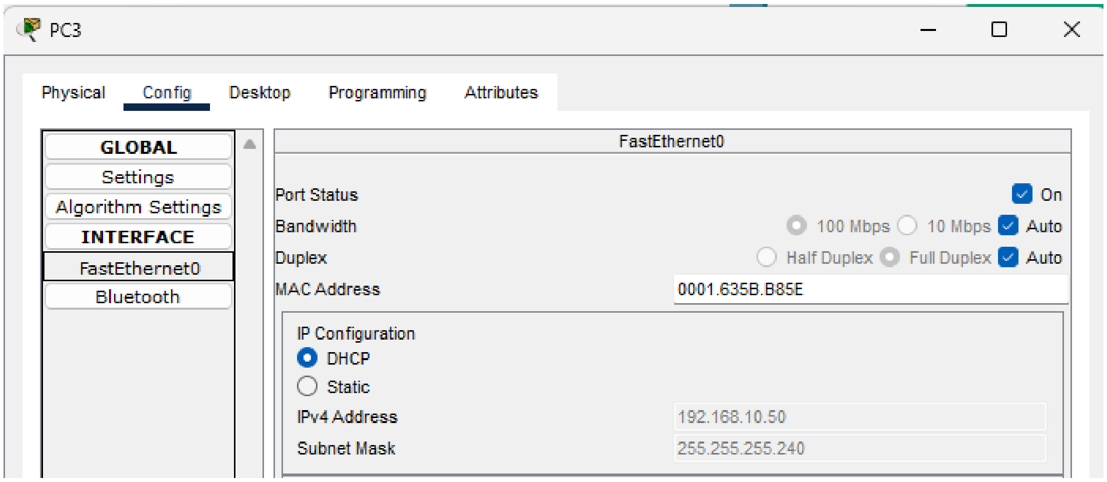


End Devices

1. IT DEPARTMENT:

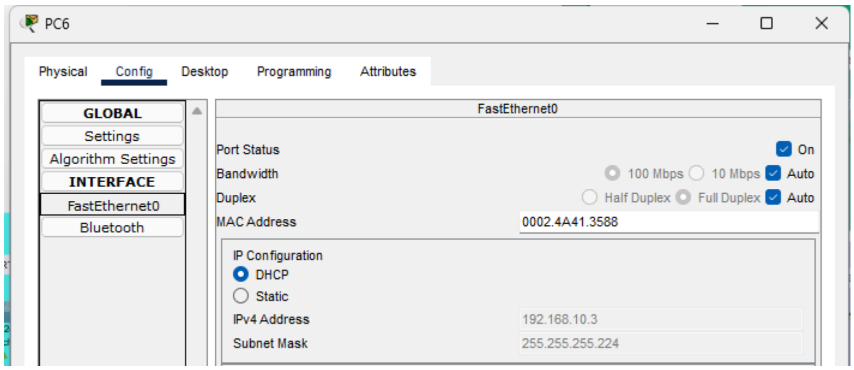
- PC2 - Connected via fa0/2 of ITDEPT_SWITCH:
- PC3 - Connected via fa0/3 of ITDEPT_SWITCH:
- **Configuration Screenshots:**

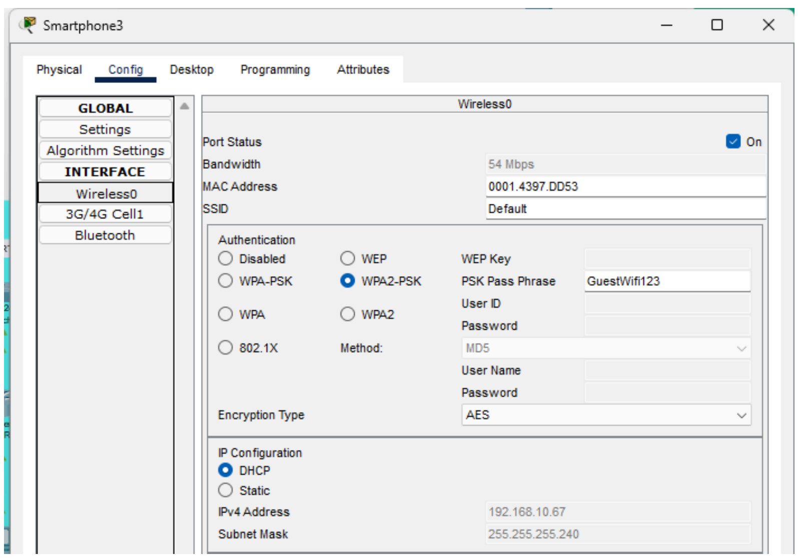


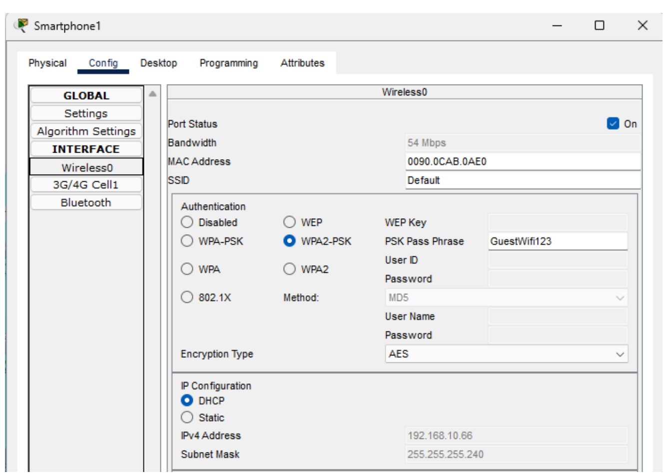


2. PHYSICAL HELPDESK ROOM:

- PC6 - Connected via fa0/3 of HELPDESK_SWITCH:
- Smartphone 1 & 3 - Connected wirelessly via access point 1:
- **Configuration Screenshots:**

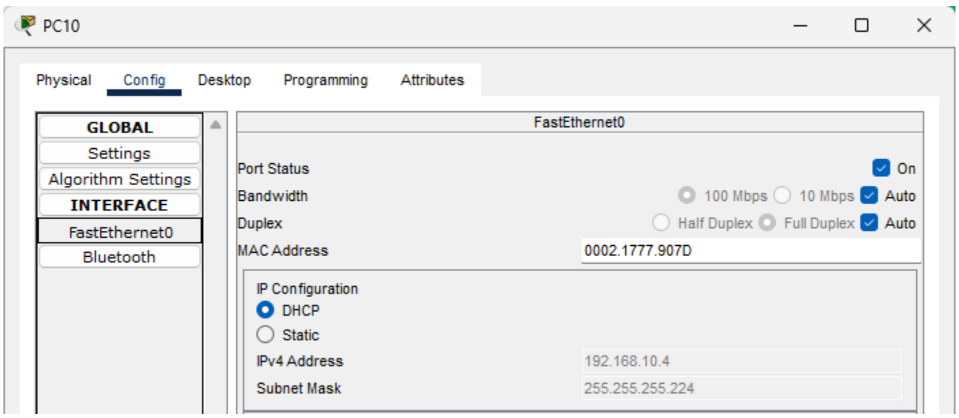


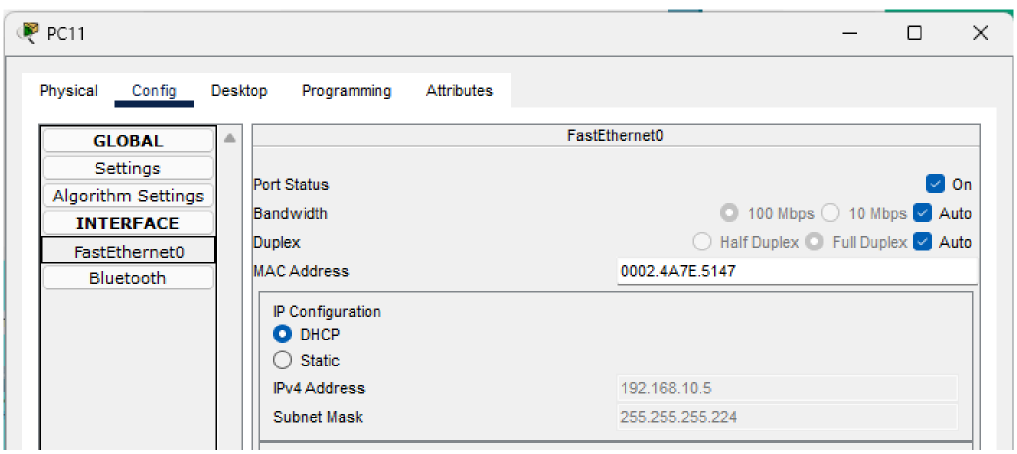


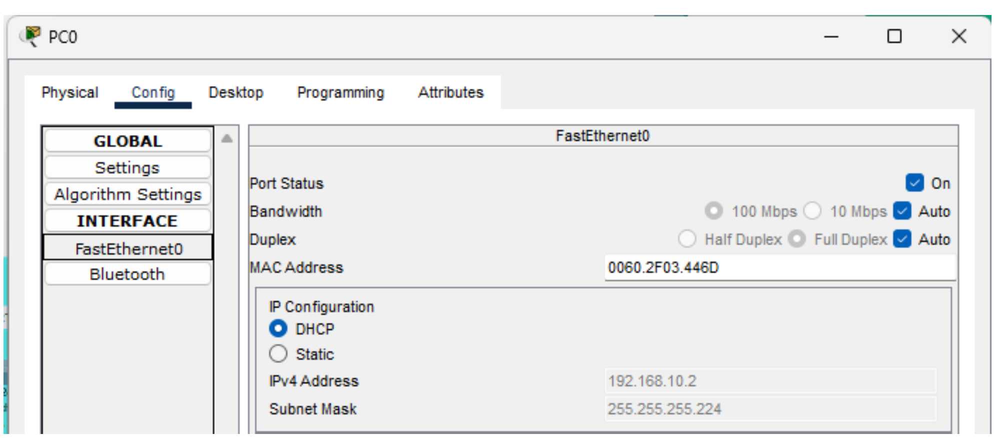


3. COMPUTER LABORATORY ROOM:

- PC10 - Connected via fa0/3 of COMLAB_SWITCH:
- PC11 - Connected via fa0/4 of COMLAB_SWITCH:
- PC0 - Connected via fa0/5 of COMLAB_SWITCH:
- Smartphone 2 - Connected wirelessly via access point 2:
- **Configuration Screenshots:**

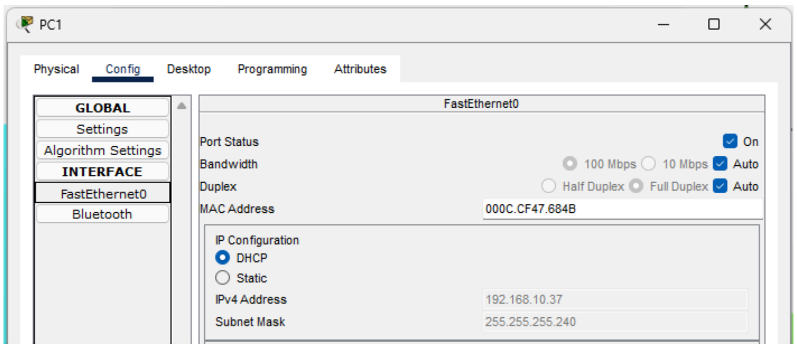


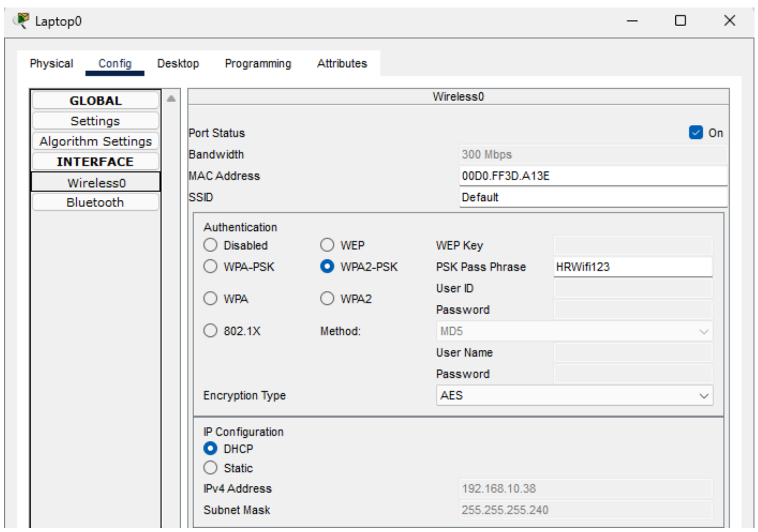


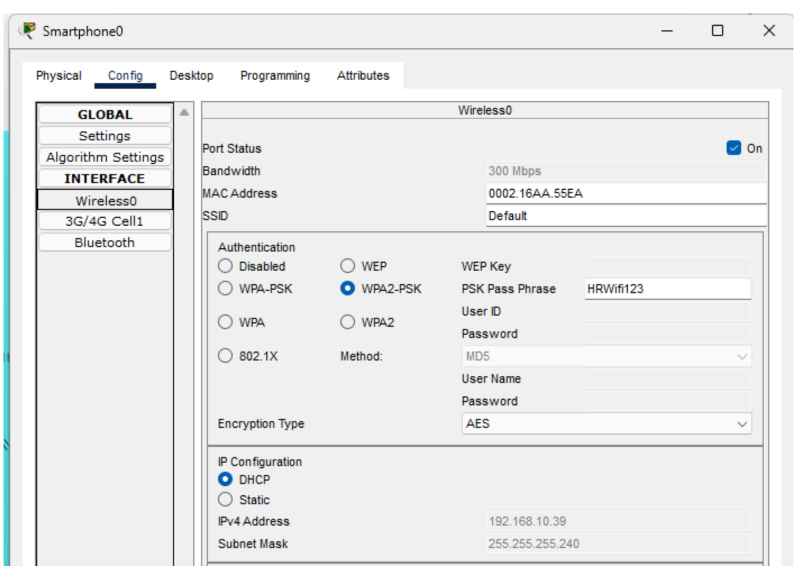


4. HR DEPARTMENT:

- PC1 - Connected via g0/1 of HOME ROUTER PT-AC:
- Laptop 0 - Connected wirelessly via HOME ROUTER PT-AC:
- Smartphone 0 - Connected wirelessly via HOME ROUTER PT-AC:
- **Configuration Screenshots:**





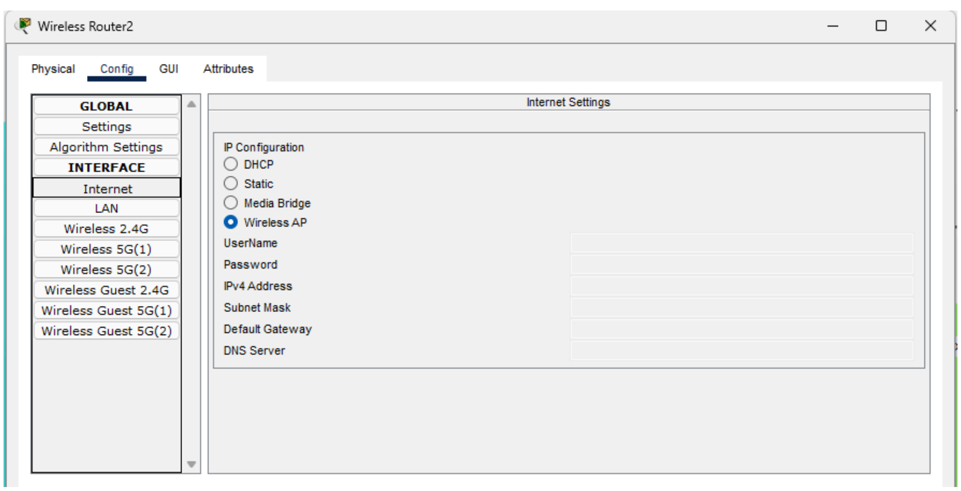


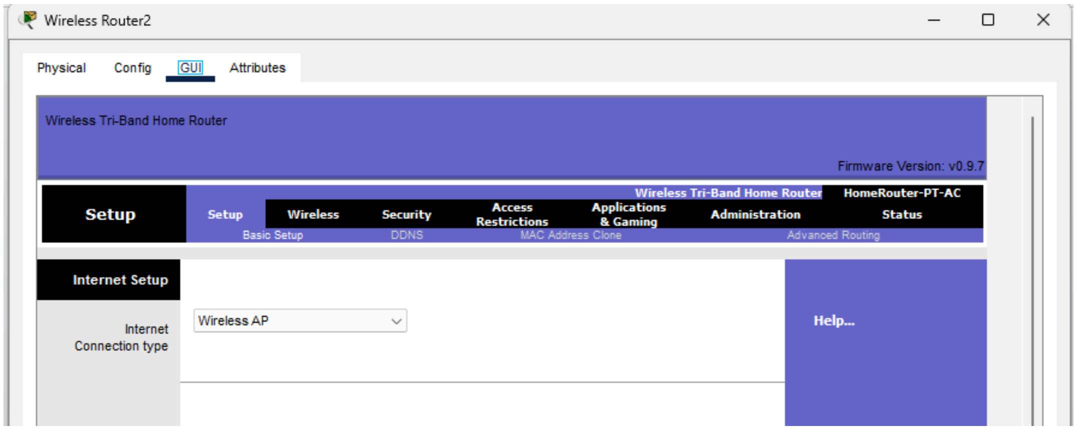
Home Router (PT-AC) Configuration

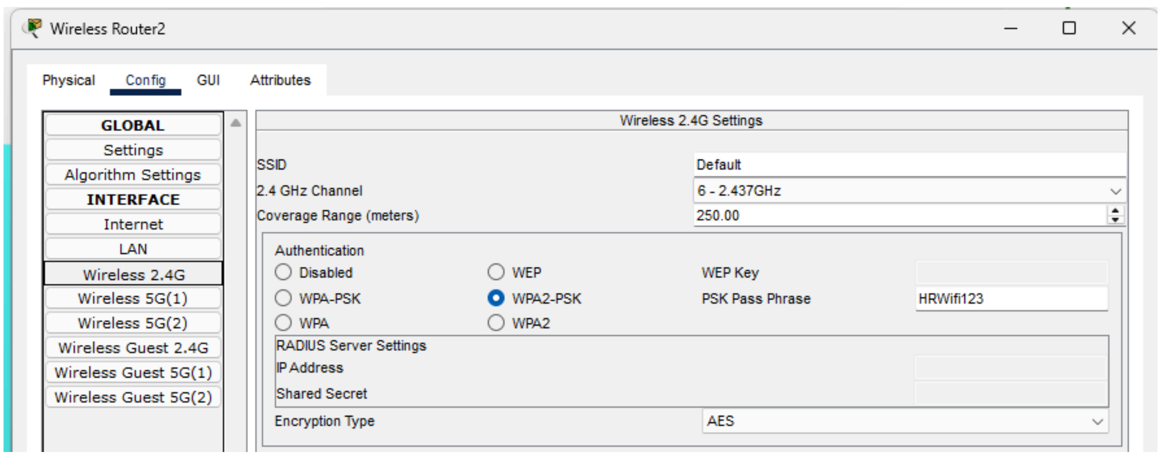
Purpose: By default, home routers have their own internal configuration for its default IP address and its own DHCP Pool. Since the goal of this laboratory is to showcase the DHCP Relay, the home router must be configured to become a working access point instead of working as a full home router.

Configurations made:

- Turned off the DHCP server and set the home router as a wireless access point.
- Instead of plugging in the fa0/2 of the HR_SWITCH directly to the WAN port of the home router PT-AC, it is plugged into the g0/2 port of the home router.
    - **Explanation:** WAN ports and LAN ports have different MAC addresses based on their network card interface (NIC).
    - **Security Mitigation:** We want to make sure that the MAC address of the LAN port is the one being learned by our HR_SWITCH, not the WAN port.
- **Configuration Screenshot:**




Wireless Router2
Physical
Config
GUI
Attributes
Wireless Tri-Band Home Router
Firmware Version: v0.9.7
Wireless Tri-Band Home Router
HomeRouter-PT-AC
Setup
Setup
Wireless
Security
Access Restrictions
Applications & Gaming
Administration
Status
Basic Setup
DDNS
MAC Address Clone
Advanced Routing
Internet Setup
Internet Connection type
Wireless AP
Help...


Wireless Router2
Physical
Config
GUI
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
Internet
LAN
Wireless 2.4G
Wireless 5G(1)
Wireless 5G(2)
Wireless Guest 2.4G
Wireless Guest 5G(1)
Wireless Guest 5G(2)
Wireless 2.4G Settings
SSID
Default
2.4 GHz Channel
6 - 2.437GHz
Coverage Range (meters)
250.00
Authentication
Disabled
WEP
WEP Key
WPA-PSK
WPA2-PSK
PSK Pass Phrase
HRWifi123
WPA
WPA2
RADIUS Server Settings
IP Address
Shared Secret
Encryption Type
AES

BREAK/FIX SCENARIO

Provisioning of home router as an AP to the HR Department

- **Identification:** Newly bought unconfigured home router.
- **Root Cause:** A consumer router (PT-AC) placed in HR was running active DHCP/NAT out of the box, conflicting with Server0.
- **Fix Applied:** Disabled local DHCP/NAT on the home router, bypassed the WAN port, connected via LAN port G0/2 to convert it into a pure Layer 2 Access Point, and adjusted Layer 2 Port Security policies on HR_SWITCH (Fa0/2) to allow bridged wireless client traffic.
- **Further explanations with provided screenshots:**

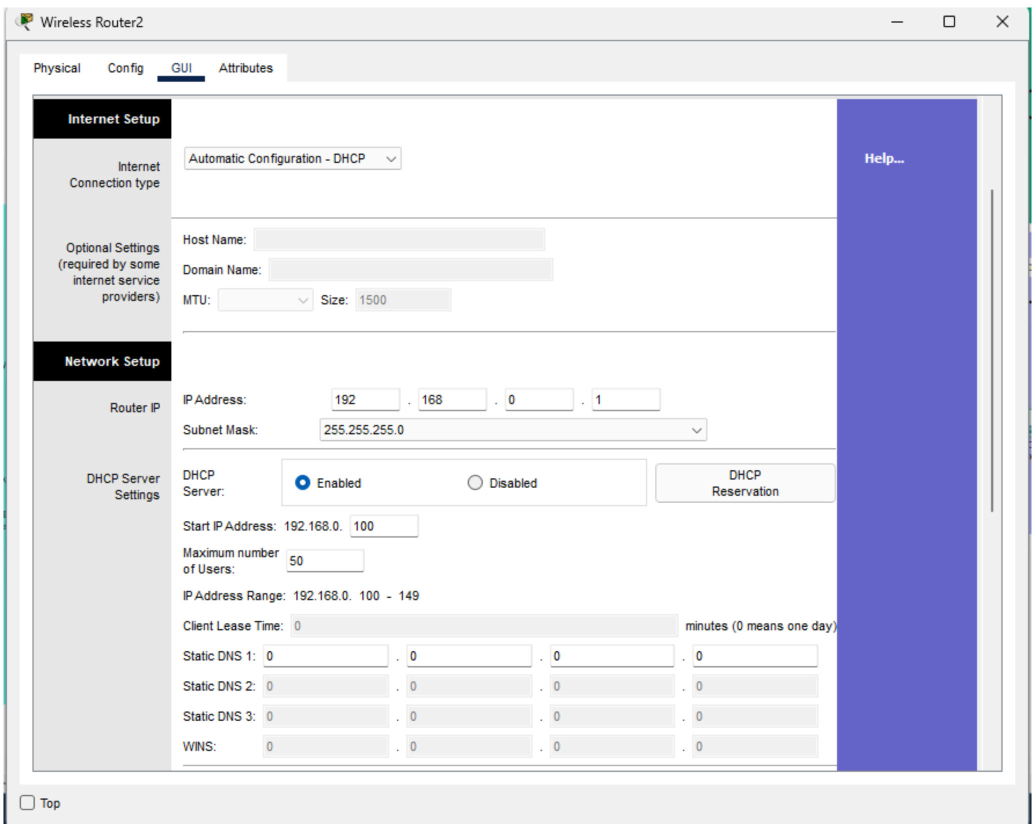


Figure 2: Unconfigured Home Router

Based on the given screenshot, the home router has an IP address of 192.168.0.1 with a subnet of /24. It also has an active pre-configured DHCP Server resulting in conflicts with the centralized server in the network topology.

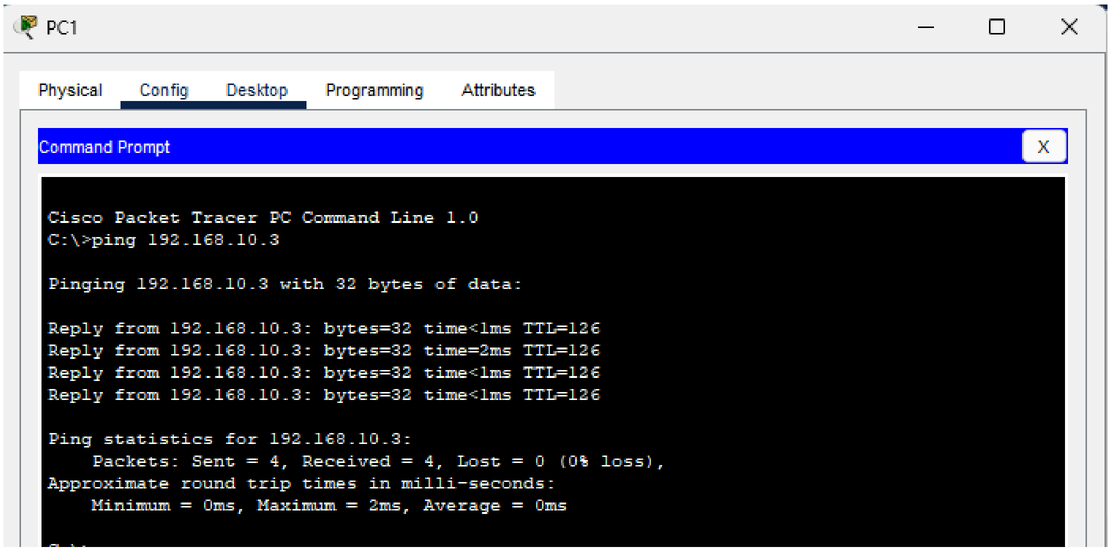


Figure 3: PC10 inside VLAN 10 reachable by PC1 inside the home router

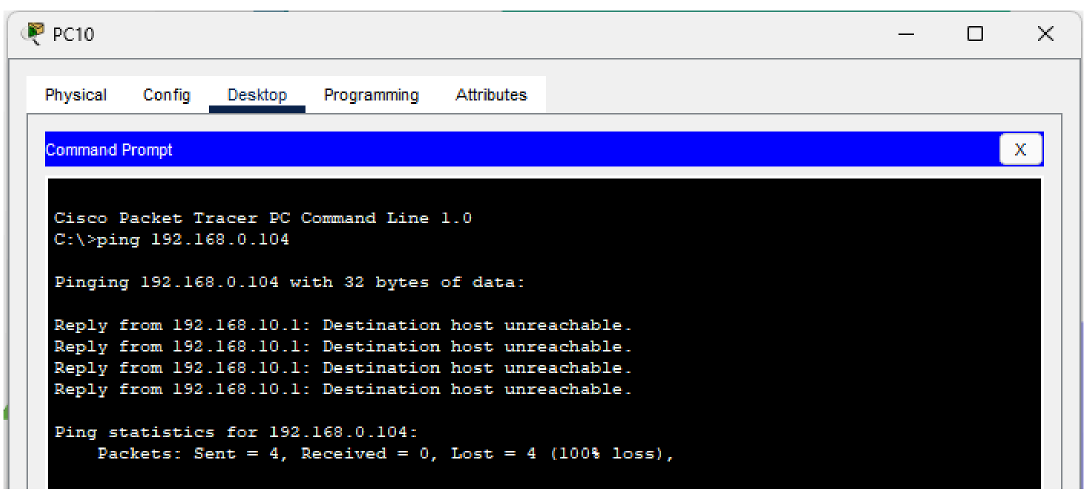


Figure 4: PC10 of VLAN 10 cannot reach the PC1 of PC1 inside the home router

Additionally, PC1 inside the home router can reach PC10 inside the VLAN 10 without a problem. While on the other hand, PC10 inside the VLAN 10 cannot reach PC1 inside the home router possibly due to its built-in firewall denying outside traffic and its own NAT inside.

- **Fixed solution:**
    1. Turned off the DHCP server and set the home router as a wireless access point.
    2. The trunk port of the HR_SWITCH providing VLAN 20 addresses was plugged in the LAN port instead of plugging it in the WAN port.
    3. The home router PT-AC is now acting as a layer 2 access point, providing valid IP addresses from the centralized server to its end devices inside the HR Department..